

Systemic Therapy in Patients With Metastatic Castration-Resistant Prostate Cancer: ASCO Guideline Clinical Insights

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Prostate cancer is a major cause of disease burden, with an estimated approximately 313,780 new cases (the highest incidence, 29% of all new cases) and approximately 35,770 deaths (the second highest cause of mortality in males and constituting 11% of all cancer-related deaths) in 2025.¹ In 2014, ASCO published recommendations in the metastatic castrate-resistant prostate cancer (mCRPC) setting,² which underwent a rapid update in 2023³ and again 2024.⁴ To account for new evidence and the content of the rapid updates, in 2025, ASCO updated the guideline to better inform clinicians, patients, and other stakeholders with up-to-date recommendations.⁵ Nearly 20%-30% of patients with mCRPC have genetic alterations of the homologous recombination repair (HRR) pathways, and the availability of poly (ADP-ribose) polymerase (PARP) inhibitors (PARPi) has improved clinical outcomes in a subset of these patients. The updated 2025 guideline addresses the following overarching questions: (1) What are the treatment options for patients with mCRPC? (2) What are the treatment options for de novo or treatment-emergent small cell neuroendocrine carcinoma of the prostate? and (3) How to assess for response while on systemic therapy for mCRPC? In this summary, we highlight the role of PARP inhibitors and immune checkpoint inhibitors (ICIs) in treatment of mCRPC. See [Figure 1](#) for a summary of key guideline recommendations, [Figure 2](#) for an algorithm for patients previously treated with androgen-deprivation therapy (ADT) alone, and [Figure 3](#) for an algorithm for patients previously treated with ADT, androgen receptor pathway inhibitor (ARPI), and docetaxel.



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WHICH PATIENTS WITH HRR ALTERATIONS ARE LIKELY TO BENEFIT WITH PARPi + ARPI COMBINATIONS?

Alterations in one of the HRR genes are commonly seen in 20%-30% of patients with mCRPC. Nearly 40%-50% of these HRR alterations are inherited. Specifically, *BRCA2* and *BRCA1* are the most common HRR alterations, followed by *ATM*, *CDK12*, *CHEK2*, and *CHEK1* mutations. The Panel recommends somatic HRR testing at the earliest opportunity to determine the presence of HRR alterations in all patients with mCRPC. Patients with de novo metastatic prostate cancer or high variant allele frequency (>50%) on somatic testing should be referred for germline testing. Patients with *BRCA1/2* alterations had a statistically significant improvement in radiographic progression-free survival (rPFS; 11.2 v 6.4 months, hazard ratio [HR], 0.5 [95% CI, 0.36 to 0.69]; $P < .001$) with single-agent rucaparib in TRITON3. In Cohort A (*BRCA1/2* and *ATM* mutations) of the PROFOUND study with single-agent olaparib, the rPFS was 7.4 v 3.6 months (HR, 0.34 [95% CI, 0.24 to 0.47]; $P < .001$). Nearly 21%-23% of patients in TRITON-3 and 44%-45% patients in PROFOUND had received previous docetaxel chemotherapy. Three large landmark studies evaluated the combination of PARPi with ARPI in the frontline mCRPC setting, PROpel (olaparib with abiraterone), MAGNITUDE (niraparib with abiraterone), and TALAPRO-2 (talazoparib with enzalutamide). HRs for rPFS were significantly better for patients with *BRCA* alterations for all three studies, HR, 0.23 (95% CI, 0.12 to 0.47) in PROpel; HR, 0.55 (95% CI, 0.39 to 0.78, $P < .0007$) in MAGNITUDE; and HR, 0.20 (95% CI, 0.11 to 0.36; $P < .00021$) in TALAPRO-2. The efficacy of PARP inhibitors varies by genetic alteration, with the greatest clinical benefit observed in patients harboring *BRCA1*, *BRCA2*, *CDK12*, and *PALB2* mutations.⁶ For patients with *BRCA1/2* alterations and previously treated with ADT alone or ADT with docetaxel for hormone-sensitive prostate cancer (HSPC), the Expert Panel recommends the use of talazoparib with enzalutamide, olaparib with

Recommendation	Evidence Quality	Strength
For patients previously treated with ADT alone in castration-sensitive or nonmetastatic CRPC setting and whose disease has progressed to metastatic CRPC, the panel recommends (Fig 2: Algorithm for patients previously treated with ADT alone)		
1.1.1. The Panel recommends HRR testing prior to initiation of systemic therapy (see the Germline and Somatic Genomic Testing for Metastatic Prostate Cancer: ASCO Guideline). ¹⁵ If positive for <i>BRCA1/2</i> alterations, then preferred options include either of niraparib + abiraterone, olaparib + abiraterone, or talazoparib + enzalutamide.	H	S
<i>Practical Information for Recommendation 1.1.1:</i> The Panel recommends personalized treatment based on therapies received during castration-sensitive setting, while also accounting for each patient's clinical status, therapy-related toxicities, potential drug interactions, cost, and drug availability as per Principles of Practice Recommendation 3. ⁵		
1.1.2. The Panel suggests the combination of talazoparib and enzalutamide for patients with any of the other HRR alterations (<i>PALB2</i> , <i>ATM</i> , <i>ATR</i> , <i>CHEK2</i> , <i>FANCA</i> , <i>RAD51C</i> , <i>NBN</i> , <i>MLH1</i> , <i>MRE11A</i> , <i>CDK12</i>).	L	C
<i>Practical Information for Recommendation 1.1.2:</i> Responses vary according to individual genes, but the strongest responses are seen in patients with <i>PALB2</i> and <i>CDK12</i> alterations in post hoc subgroup analyses.		
For patients previously treated with ADT, ARPI, and docetaxel, and have progressive metastatic CRPC (Fig 3: Algorithm for patients previously treated with ADT, ARPI, and docetaxel).		
1.4.2. The Panel recommends olaparib monotherapy for patients with HRR alterations.	M	S
<i>Practical Information for Recommendation 1.4.2:</i> As it has demonstrated a PFS benefit, rucaparib may be offered as an alternative to olaparib for selected patients with <i>BRCA1/2</i> alterations based on clinical judgment and patient preferences, pending availability.		

FIG 1. Summary of key guideline recommendations. ADT, androgen deprivation therapy; ARPI, androgen receptor pathway inhibitor; C, conditional; CRPC, castration-resistant prostate cancer; HRR, homologous recombination repair; L, low; M, moderate; PFS, progression-free survival; S, strong.

abiraterone, or niraparib with abiraterone. However, there are no head-to-head trials directly comparing the safety and efficacy of these regimens. The choice of combination should be guided by factors such as drug availability, cost, and potential drug interactions. While the toxicity profiles of PARP inhibitors are generally similar, the side effect profiles of abiraterone and enzalutamide differ and should be taken into account when selecting a treatment regimen. For patients with *BRCA1/2* alterations who received doublet (ADT + ARPI) or triplet therapy (ADT + ARPI + docetaxel) for HSPC, either olaparib or rucaparib is recommended. For patients with non-*BRCA1/2* HRR alterations, talazoparib with enzalutamide (for those who are ARPI-naïve) or olaparib (for those previously treated with an ARPI) is recommended.

FOR PATIENTS WITH HRR ALTERATIONS AND PSMA-POSITIVE mCRPC, HOW DO YOU DECIDE BETWEEN PARP INHIBITORS AND ¹⁷⁷Lu-PSMA-617?

Currently, there are no randomized clinical trials directly comparing PARP inhibitors and radioligand therapies in patients with HRR alterations and prostate-specific membrane antigen (PSMA)-positive mCRPC. Retrospective analyses suggest

that patients with *BRCA* alterations treated with ¹⁷⁷Lu-PSMA-617 may experience inferior PFS and overall survival compared with those without such alterations.⁷ These findings require validation in larger prospective cohorts. In addition, the TRITON-3 trial demonstrated that rucaparib significantly improved PFS compared with docetaxel in patients with *BRCA*-mutated mCRPC (11.2 v. 8.3 months; HR, 0.53 [95% CI, 0.37 to 0.77]).⁸ These findings support the use of PARP inhibitors earlier in the treatment before ¹⁷⁷Lu-PSMA-617 or chemotherapy for patients with *BRCA* alterations. However, data regarding the treatment sequencing for patients with non-*BRCA* HRR alterations require additional randomized data.

WHAT ARE SOME STRATEGIES THAT COULD BE USED TO ADDRESS TOXICITIES WITH PARPI?

The most common side effects of PARP inhibitors include myelosuppression, fatigue, nausea, vomiting, and decreased appetite. These typically occur within the first 3 months of treatment and hence need regular monitoring with blood work. For any grade 3 or 4 hematologic toxicities (hemoglobin <8.0 g/dL, ANC <1,000/μL, or platelets <50,000/μL), PARP inhibitors should be withheld. Transfusions for severe

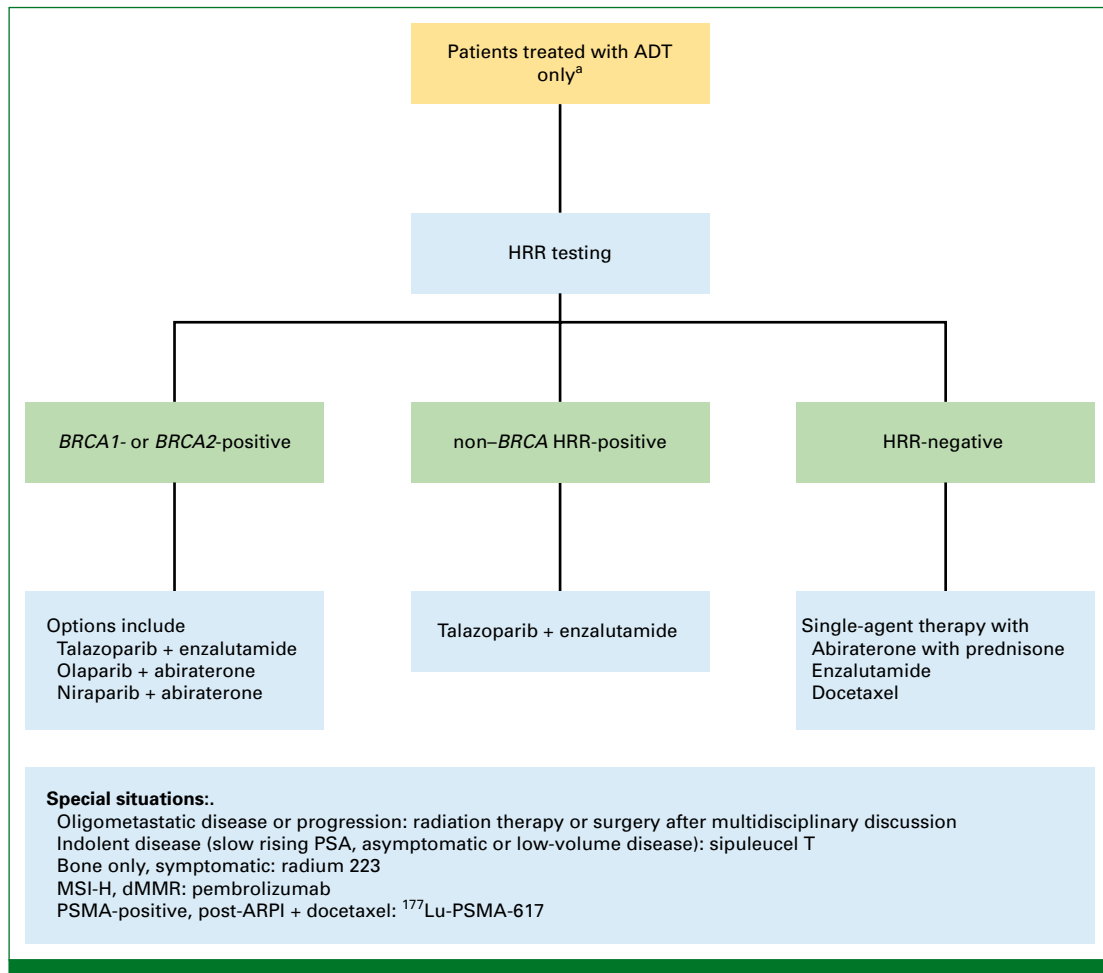


FIG 2. Algorithm for patients previously treated with ADT alone. ¹⁷⁷Lu-PSMA-617, lutetium-177-PSMA-617; ADT, androgen deprivation therapy; dMMR, mismatch repair-deficient; HRR, homologous recombination repair; MSI-H, microsatellite instability-high; PSA, prostate-specific antigen; PSMA, prostate-specific membrane antigen. ^aOptions listed are preferred options.

anemia or thrombocytopenia should follow institutional protocols. The use of granulocyte colony-stimulating factors (G-CSF) is recommended only in cases of febrile neutropenia; primary prophylaxis with G-CSF is not advised. Once toxicities improve, treatment should be reinitiated at a reduced dose. For other nonsevere but symptomatic toxicities, dose reduction should also be considered. Although the risk of myelodysplastic syndrome and acute myeloid leukemia is rare, patients with persistent myelosuppression despite discontinuing PARP inhibitors and receiving appropriate supportive care should be promptly evaluated for these conditions.

WHAT IS THE ROLE OF ICIs IN PATIENTS WITH mCRPC?

Although several prospective studies have evaluated ICI in unselected patients with mCRPC, their clinical benefit

has largely been confined to those with microsatellite instability-high (MSI-H) or mismatch repair deficiency (dMMR) and tumor mutation burden (TMB) ≥ 10 mut/Mb. Currently, pembrolizumab and dostarlimab are US Food and Drug Administration (FDA)-approved for broader tumor-agnostic use in patients with MSI-H/dMMR, regardless of cancer type.^{9,10} In addition, pembrolizumab has received approval for tumors with TMB ≥ 10 mut/Mb.¹⁰ However, it is important to note that only a small number of patients with prostate cancer were included in these trials that led to FDA approval. Single arm studies in patients with MSI-H/dMMR tumors showed greater prostate-specific antigen (PSA) 50% decline and objective response with ICI.^{11,12} While MSI-H tumors are generally associated with high TMB, tumors characterized by high TMB but microsatellite stability (MSS) exhibit heterogeneous biology and have been associated with lower response rates to ICI.^{11,13,14}

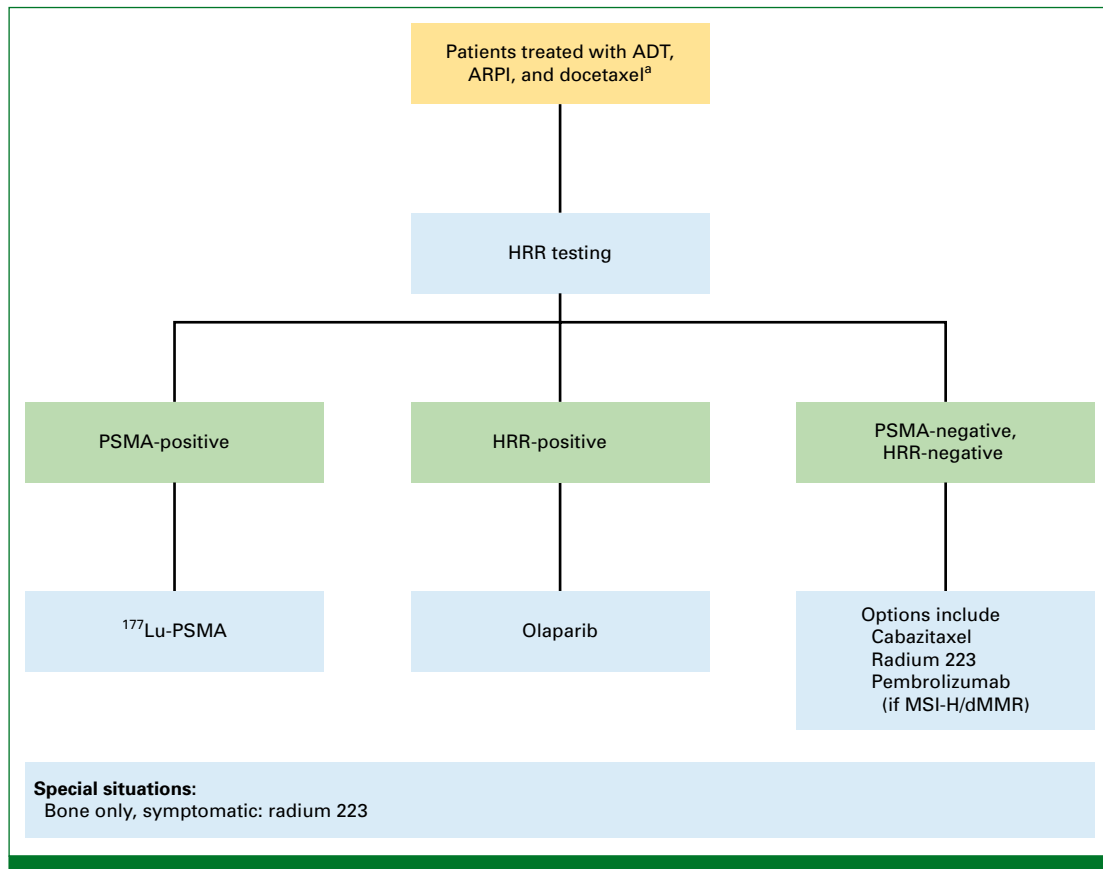


FIG 3. Algorithm for patients previously treated with ADT, ARPI, and docetaxel. ¹⁷⁷Lu-PSMA-617, lutetium-177-PSMA-617; ADT, androgen deprivation therapy; ARPI, androgen receptor pathway inhibitor; dMMR, mismatch repair–deficient; HRR, homologous recombinant repair; MSI-H, microsatellite instability-high; PSA, prostate-specific antigen; PSMA, prostate-specific membrane antigen. ^aOptions listed are preferred options.

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